Drew University - CSCI 260 - Fall 2024
In-Class Exercise 2 - Digital Logic

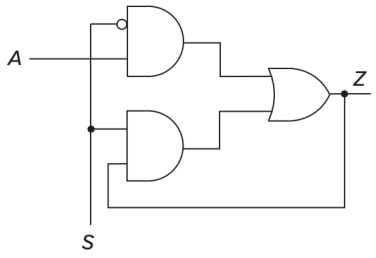
Instruction: Please work in groups of 4 and write down your names for attendance. 09/20/2024
Names:

1. For the logic expression NOT(C OR NOT(A)) AND (C OR B), first give its equivalent logic circuit and then use a truth table to evaluate it.

2. Show that the NAND gate, by itself, is logically complete. To do so, construct logical circuits (i.e., diagrams) that perform the NOT, AND, and OR operations by only using the NAND gate.

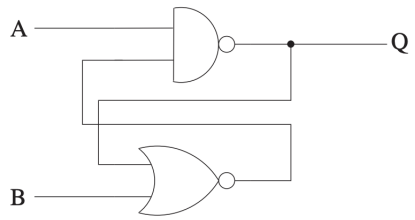
3. For the following logic circuit below, answer the following questions.

(a) Describe its output Z using a truth table when the select signal S is a logical 0.

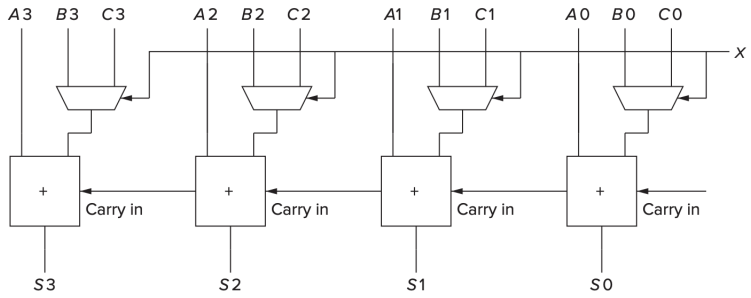


(b) If the select signal S is switched from a 0 to 1, what will the output be?

4. Given the latch below, for what values of A and B will the latch remain in the quiescent/quiet state (i.e., the output will not change)?



5. The figure below shows a logic circuit that appears in many of today's processors. Each of the boxes is a full-adder circuit and each of the trapezoids is a 2-to-1 MUX. What does the value on the wire X do?



6. Implement a 4-to-1 MUX using only 2-to-1 MUXes. You can treat the 2-to-1 MUX as a blackbox (i.e., no need to show the gates), but when implementing, make sure you connect all the necessary wires. Also, write out the truth table of the 4-to-1 MUX for selecting input. **Show all your work**.

   Remember that a 4-to-1 MUX has 4 input sources, 2 control signals, and 1 output. (Hint: How do we make a 3-input AND gate out of 2-input AND gates?)